

(20) When $b = 0$, it is often stated and used in the contrapositive form.

21. $Ax = 0$ has a non-zero solution $\Rightarrow |A| = 0$.

The contrapositive of a statement $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$. The two statements say the same thing.

Theorem. Let A be an $n \times n$ matrix.

22. $|A| = 0 \Rightarrow Ax = 0$ has a non-trivial (i.e. non-zero) solution.

23. $|A| = 0 \Rightarrow Ax = b$ usually has no solution, but has solution for some b .

We call the system *consistent* if it has solutions; otherwise it is inconsistent. This probably seems like a maze of similar-sounding and confusing theorems. Let's get another perspective on these ideas by seeing how they apply separately to homogeneous and inhomogeneous systems.

Homogeneous systems:

$$Ax = 0 \text{ has non-trivial solutions } \iff |A| = 0.$$

Inhomogeneous systems:

$$Ax = b \text{ has a unique solution } x = A^{-1}b, \text{ if } |A| \neq 0.$$

If $|A| = 0$, then $Ax = b$ usually has no solutions, but it does have solutions for some b .

The statements (20) and (21) are proven since we have a formula for the solution, and it is easy to see by multiplying

$$Ax = b \text{ by } A^{-1}.$$

If x is the solution, then

$$x = A^{-1}b.$$

Let A have three rows given by the row vectors a, b, c , and let $x = (x, y, z)$. Then

$$Ax = 0$$

is the same system as

$$a \cdot x = 0, \quad b \cdot x = 0, \quad c \cdot x = 0.$$

Homogeneous and Inhomogeneous

$$|A| = a \cdot (b \times c)$$

= volume of the parallelepiped determined by a, b, c .

It follows that if $|A| = 0$, the parallelepiped has zero volume, and therefore the three vectors a, b, c lie in a plane.

Any non-zero vector x which is orthogonal to the plane will then be orthogonal to a, b, c .

If $|A| = 0$, then $Ax = 0$ has a non-trivial solution.

Write the system

$$Ax = d$$

where d is the column vector

$$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}.$$

Writing this out (as in question 24), it becomes

$$a \cdot x = d_1, \quad b \cdot x = d_2, \quad c \cdot x = d_3.$$

If $|A| = 0$, the three vectors a, b, c lie in a plane, which means we can write one of them, say c , as a linear combination of a and b . If however, d does not satisfy the relation (28), then the last equation in (25) is a consequence of the first two and can be discarded, and we get a system of two equations in three unknowns, which will in general have a non-zero solution, unless they represent two planes which are parallel. Singular matrices; computational difficulties. Because so much depends on whether $|A|$ is zero or not, this property is given a name. We say the square matrix A is singular if $|A| = 0$, and nonsingular or invertible if $|A| \neq 0$. Indeed, we know that $|A| \neq 0$ exists if and only if $|A| \neq 0$ which explains the term “invertible”; the use of “singular” will be familiar to Sherlock Holmes fans: it is the 19th century version of “peculiar” or the late 20th century word “special”. Even if A is nonsingular, the solution of $Ax = b$ is likely to run into trouble if $|A| \neq 0$. or as one says, A is almost-singular. Namely, in the formula for the $|A|$ occurs in the denominator, so that unless there is some sort of compensation for this in the numerator, the solutions are likely to be very sensitive to small changes in the coefficients of A , i.e., to the coefficients of the equations. Systems (of any kind) whose solutions behave this way

HOMOGENEOUS AND INHOMOGENEOUS SYSTEMS 3 are said to be ill-conditioned; it is difficult to solve such systems numerically and special methods must be used. To see the difficulty geometrically, think of a

$$2 \times 2 \text{ system } Ax = b$$

as representing a pair of lines; the solution is the point in which they intersect. If $|A| \neq 0$, but its entries are not small, then its two rows must be vectors which are almost parallel (since they span a parallelogram of small area). The two lines are therefore almost parallel; their intersection point exists, but its position is highly sensitive to the exact positions of the two lines, i.e., to the values of the coefficients of the system of equations

$$r, s \in \mathbb{R}.$$

$$cx = r(ax) + s(bx)$$

$$d_3 = rd_1 + sd_2$$

A is singular if $|A| = 0$ and non-singular if $|A| \neq 0$.
 $|A| \neq 0$ appears in the denominator of A^{-1} .

$$2 \times 2 \text{ system } Ax = b$$

They intersect if $|A| \neq 0$.